Module 13: WLAN Configuration

Switching, Routing, and Wireless Essentials v7.0 (SRWE)

Module Objectives

Module Title: WLAN Configuration

Module Objective: Implement a WLAN using a wireless router and WLC.

Topic Title	**Topic Objective**
Remote Site WLAN Configuration	Configure a WLAN to support a remote site.
Configure a Basic WLAN on the WLC	Configure a WLC WLAN to use the management interface and WPA2 PSK authentication.
Configure a WPA2 Enterprise WLAN on the WLC	Configure a WLC WLAN to use a VLAN interface, a DHCP server, and WPA2 Enterprise authentication.
Troubleshoot WLAN Issues	Troubleshoot common wireless configuration issues.

13.1 Remote Site WLAN Configuration

Video – Configure a Wireless Network

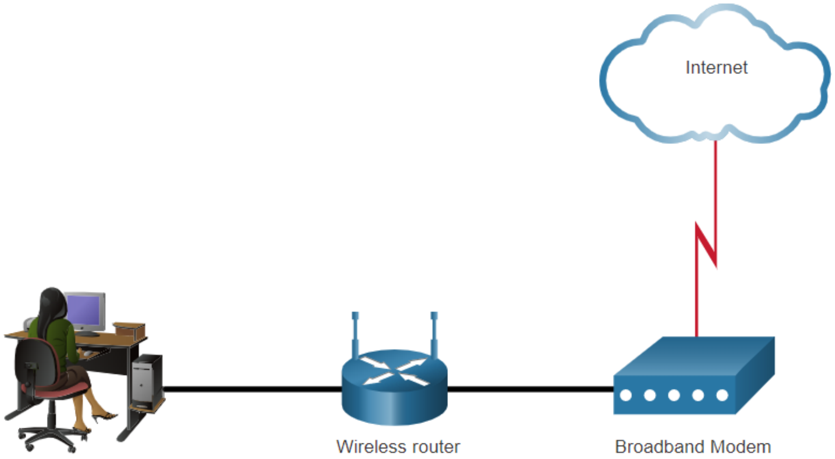
This video will cover the following:

- Use the Wireless Router Web Page
- Change the Password
- Change the WAN and LAN settings
- Connect the Wireless Network

The Wireless Router

Remote workers, small branch offices, and home networks often use a small office and home router.

- These "integrated" routers typically include a switch for wired clients, a port for an internet connection (sometimes labeled "WAN"), and wireless components for wireless client access.
- These wireless routers typically provide WLAN security, DHCP services, integrated Name Address Translation (NAT), quality of service (QoS), as well as a variety of other features.
- The feature set will vary based on the router model.



Note: Cable or DSL modem configuration is usually done by the service provider's representative either on-site or remotely.

Log in to the Wireless Router

Most wireless routers are preconfigured to be connected to the network and provide services.

- Wireless router default IP addresses, usernames, and passwords can easily be found on the internet.
- Therefore, your first priority should be to change these defaults for security reasons.

To gain access to the wireless router's configuration GUI

- Open a web browser and enter the default IP address for your wireless router.
- The default IP address can be found in the documentation that came with the wireless router or you can search the internet.
- The word **admin** is commonly used as the default username and password.

Basic Network Setup

Basic network setup includes the following steps:

- Log in to the router from a web browser.
- Change the default administrative password.
- Log in with the new administrative password.
- Change the default DHCP IPv4 addresses.
- Renew the IP address.
- Log in to the router with the new IP address.

Basic Wireless Setup

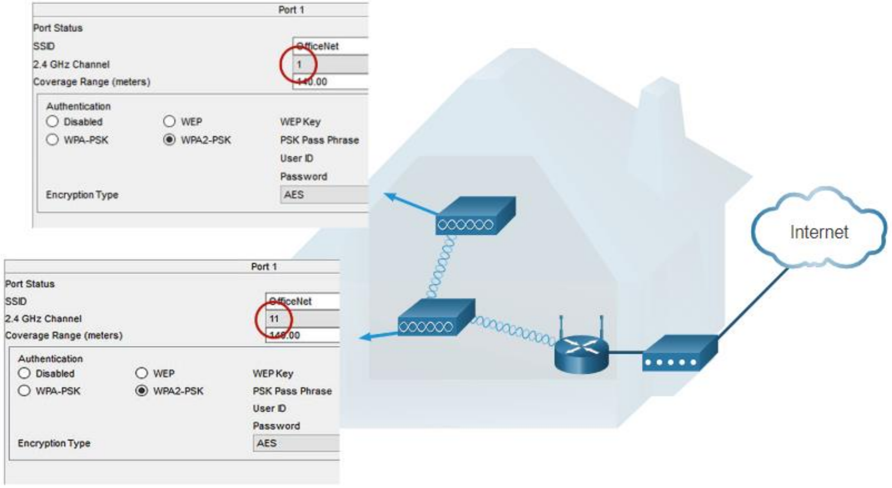
Basic wireless setup includes the following steps:

- View the WLAN defaults.
- Change the network mode, identifying which 802.11 standard is to be implemented.
- Configure the SSID.
- Configure the channel, ensuring there are no overlapping channels in use.
- Configure the security mode, selecting from Open, WPA, WPA2 Personal, WPA2 Enterprise, etc..
- Configure the passphrase, as required for the selected security mode.

Configure a Wireless Mesh Network

In a small office or home network, one wireless router may suffice to provide wireless access to all the clients.

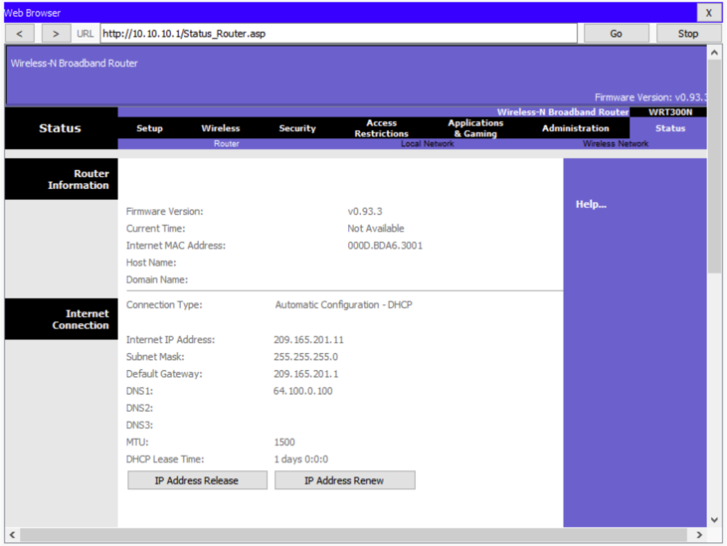
- If you want to extend the range beyond approximately 45 meters indoors and 90 meters outdoors, you create a wireless mesh.
- Create the mesh by adding access points with the same settings, except using different channels to prevent interference.
- Extending a WLAN in a small office or home has become increasingly easier.
- Manufacturers have made creating a wireless mesh network (WMN) simple through smartphone apps.



NAT for IPv4

Typically, the wireless router is assigned a publicly routable address by the ISP and uses a private network address for addressing on the LAN.

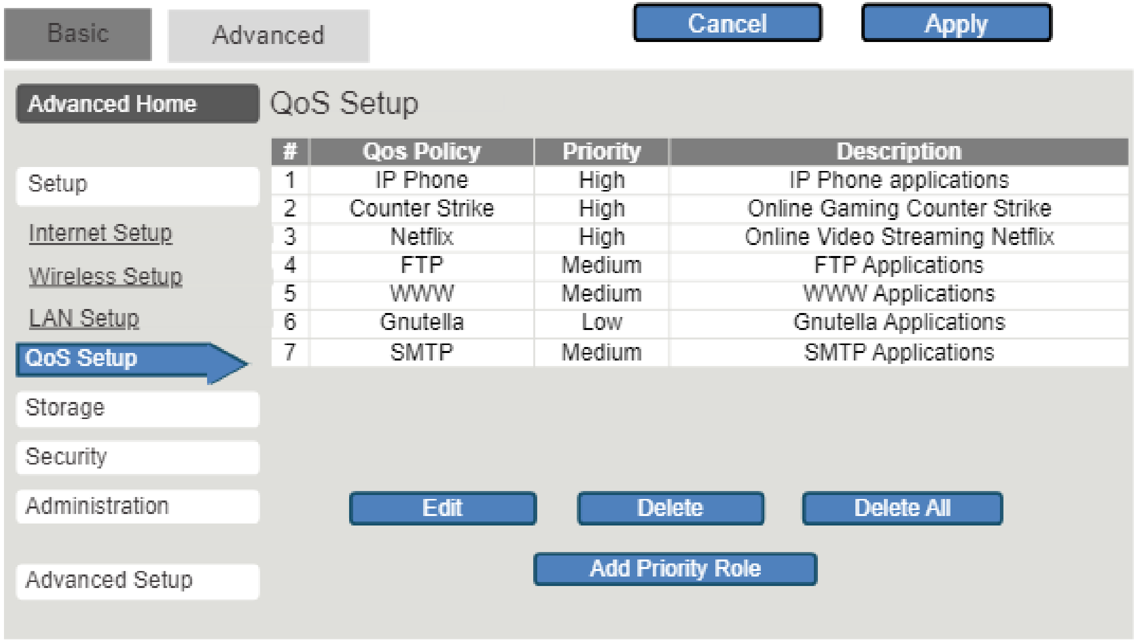
- To allow hosts on the LAN to communicate with the outside world, the router will use a process called Network Address Translation (NAT).
- NAT translates a private (local) source IPv4 address to a public (global) address (the process is reversed for incoming packets).
- NAT makes sharing one public IPv4 address possible by tracking the source port numbers for every session established by a device.
- If your ISP has IPv6 enabled, you will see a unique IPv6 address for each device.



Quality of Service

Many wireless routers have an option for configuring Quality of Service (QoS).

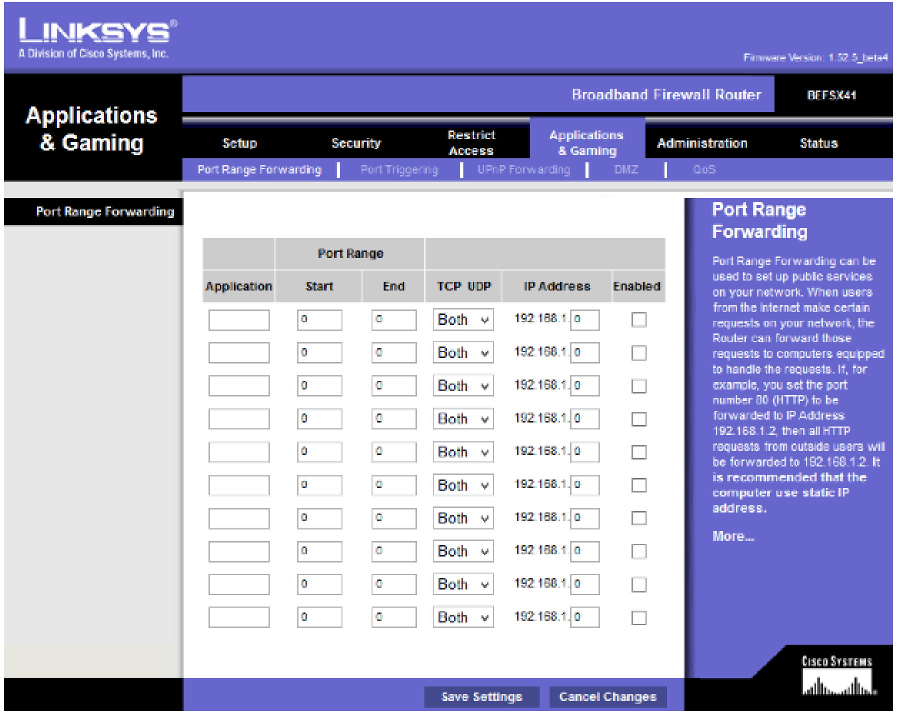
- By configuring QoS, you can guarantee that certain traffic types, such as voice and video, are prioritized over traffic that is not as time-sensitive, such as email and web browsing.
- On some wireless routers, traffic can also be prioritized on specific ports.



Port Forwarding

Wireless routers typically block TCP and UDP ports to prevent unauthorized access in and out of a LAN.

- However, there are situations when specific ports must be opened so that certain programs and applications can communicate with devices on different networks.
- Port forwarding is a rule-based method of directing traffic between devices on separate networks.
- Port triggering allows the router to temporarily forward data through inbound ports to a specific device.
- You can use port triggering to forward data to a computer only when a designated port range is used to make an outbound request.



Packet Tracer – Configure a Wireless Network

In this Packet Tracer activity, you will complete the following objectives:

- Connect to a wireless router
- Configure the wireless router
- Connect a wired device to the wireless router
- Connect a wireless device to the wireless router
- Add an AP to the network to extend wireless coverage
- Update default router settings